



Google Developer Group
On Campus

TechSprint



Leveraging the power of AI



Team Details

- a. **Team name:** Think Tank
- b. **Problem Statement:** Smart pre booking to reduce wastage of food

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Problem Statement

Understanding the impact of food wastage in college messes

- College messes prepare food based on assumptions → daily food wastage
- No advance data on student attendance → excess leftovers
- **Solution:** Pre-booking-based mess system
- Students book meals in advance via mobile/web app
- Booking data analyzed using AI to predict demand
- Mess staff cook only the required quantity
- **Outcome:** Reduced food wastage, cost savings, improved sustainability



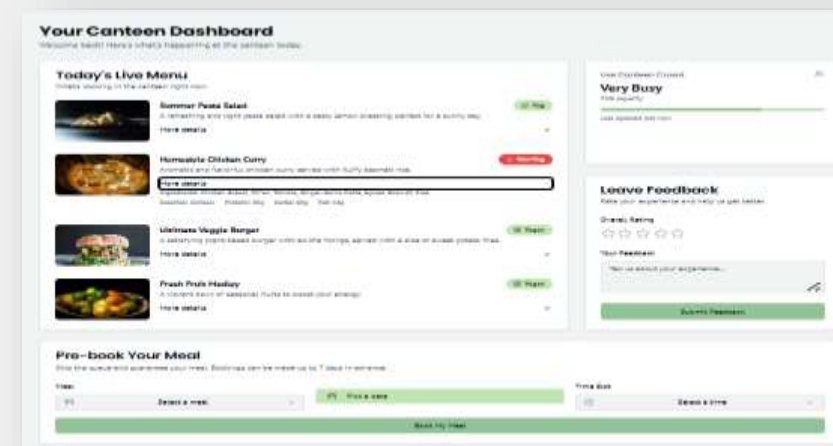
Opportunities & Innovation

How is it different?

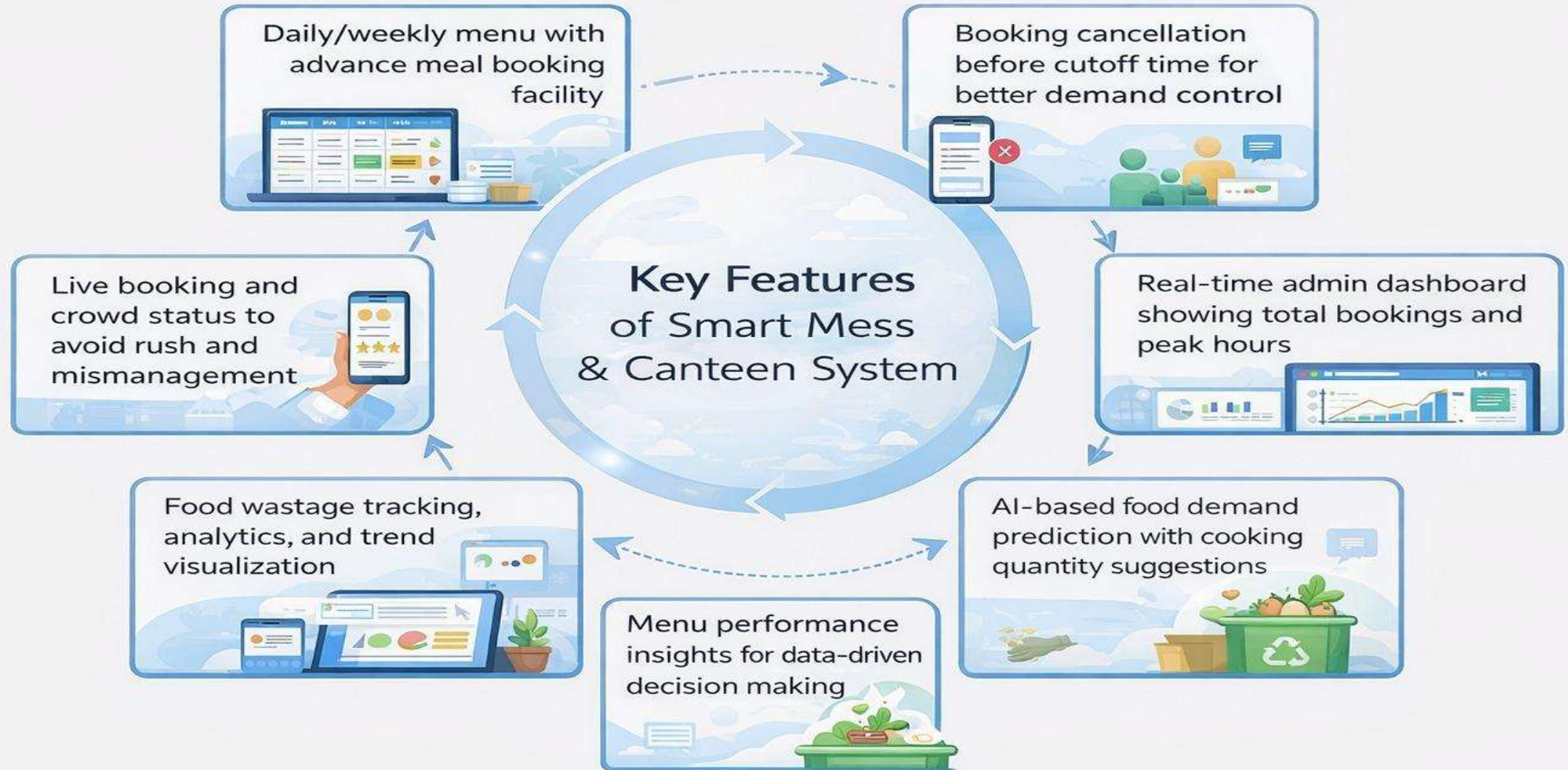
- Traditional messes rely on fixed or manual estimation
- Our system uses real-time student booking data
- AI-driven demand prediction replaces guesswork
- Focused on food wastage reduction, not just convenience
- Built on scalable Google technologies

How does it solve the problem?

- Advance booking → clear demand visibility
- AI prediction → optimal food preparation
- Cancellation window → last-minute control
- Analytics → identify recurring wastage trends
- Feedback system → better menu planning



Key Features of Smart Mess & Canteen System



Google technologies used



Firebase Authentication (Google Sign-In)

- Secure login for students and admin
- No password management required
- Restricted access to college users



Gemini API (AI)

- Predicts daily food demand from booking data
- Analyse student feedback sentiment
- Recommends optimal cooking quantities



Google Cloud Functions

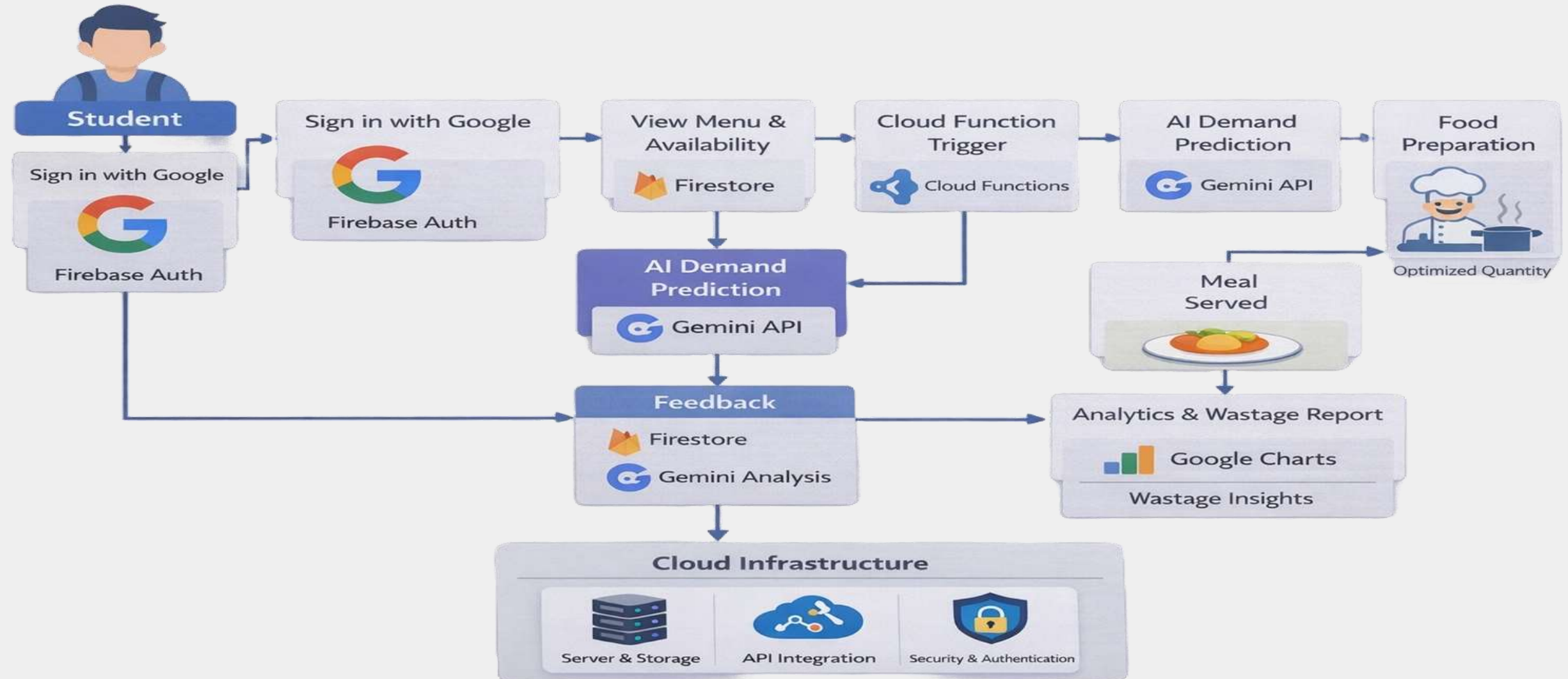
- Handles meal booking & cancellation logic
- Tracks real-time booking counts
- Runs AI demand prediction & wastage analytics



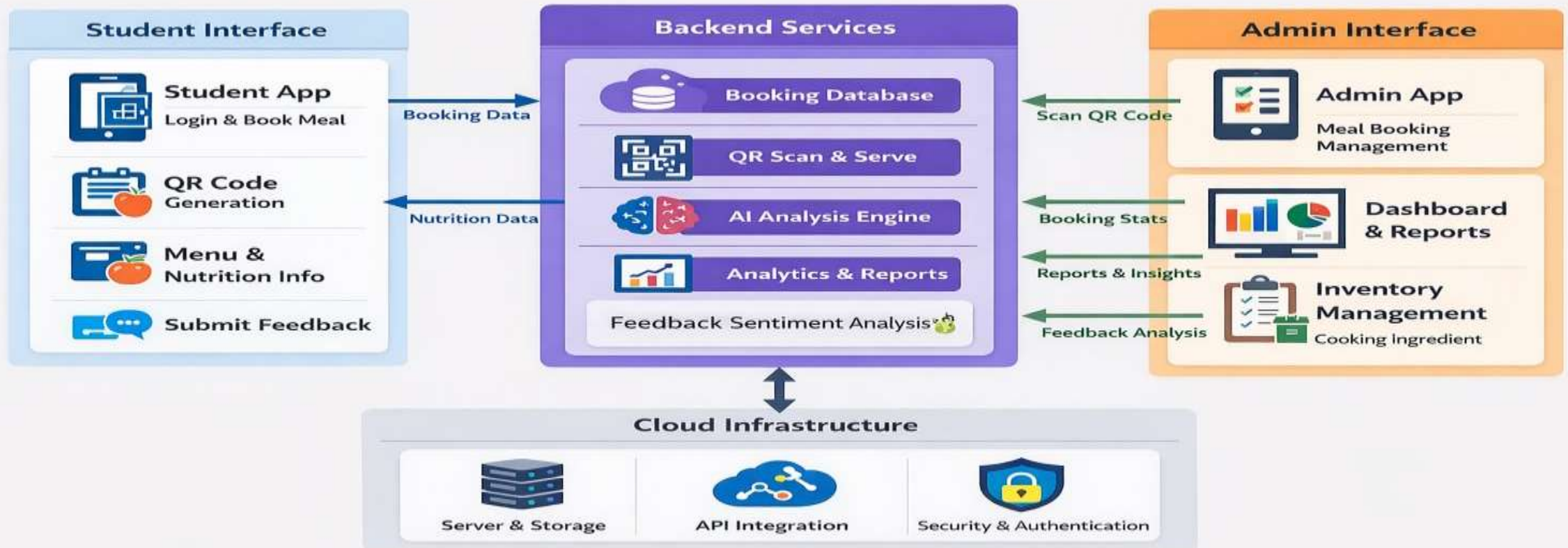
Google Charts

- Visualizes booking and demand trends
- Displays food wastage analytics
- Shows demand vs consumption graphs

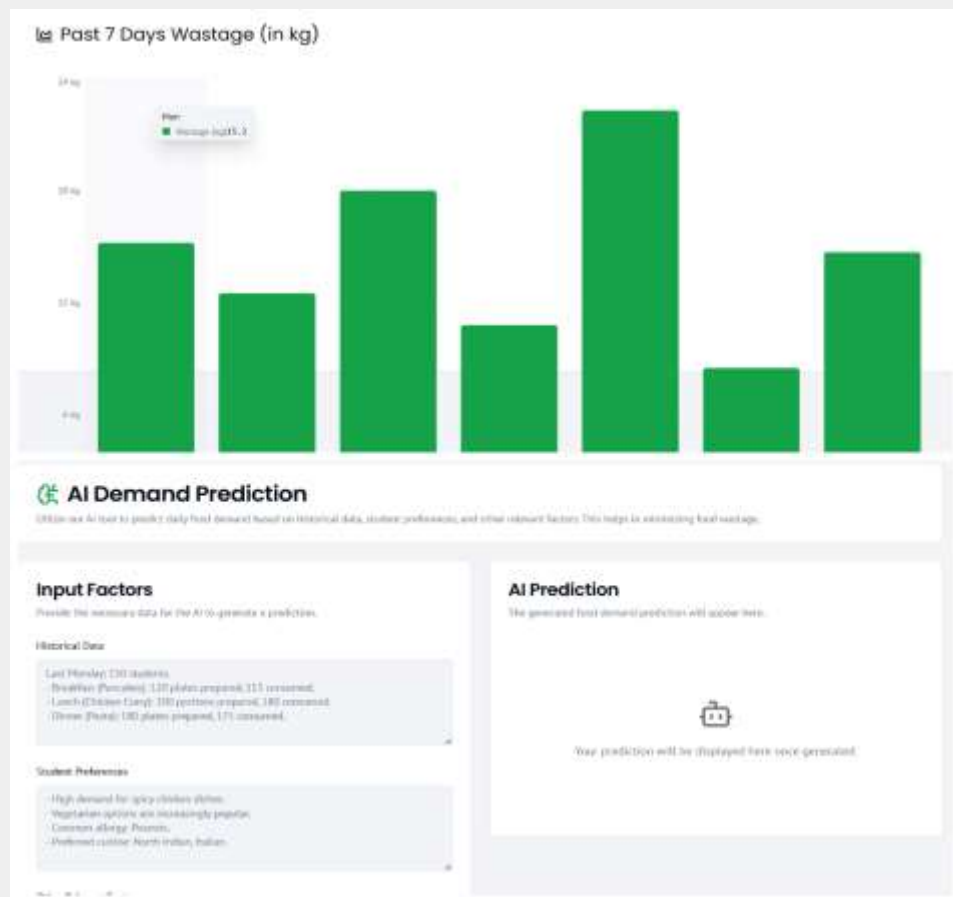
Process flow of smart mess and canteen system



Architecture diagram of the proposed solution



Snapshots of MVP



Admin dashboard

Weekly Menu
Plan your meals for the week. Click on a meal to book it.

Monday Tuesday Wednesday Thursday Friday Saturday Sunday

Breakfast (7:30 AM - 9:30 AM)

Aloo Paratha
With curd and pickle.

[Book Meal](#)

Meal Feedback
Your feedback helps us improve! Rate the meal you've had.

Rate Your Past Meals

Chole Bhature
Tuesday Lunch

[Give Feedback](#)

Welcome back, Student!
Ready to see what's cooking? Plan your meals for the day.

[View Full Menu](#)

Your Upcoming Meals

Live Crowd Status
Current bookings for today's meals.

Breakfast 42 / 110

Lunch 78 / 110

Dinner 62 / 110

My Bookings
Here are all your upcoming and past meal bookings. You can cancel upcoming meals before the cutoff time.

Your Meal Schedule

Day	Meal	Time	Status	Action
Monday	Rajma Chawal	Lunch	Booked	Cancel
Monday	Panner Bitter Masala	Dinner	Booked	Cancel
Tuesday	Chole Bhature	Lunch	Booked	Cancel

Student dashboard

Future Enhancements

Redistribution of Excess Food to NGOs

- Automated identification of surplus food after meal service using real-time consumption data
- Integration with nearby NGOs/food banks for timely pickup and distribution
- Reduces food wastage while supporting social welfare and community hunger relief



QR Code–Based Meal Redemption (Digital Ticketing)

- Generates a unique QR code after booking to verify the student at the time of meal serving
- Mess staff scan the QR to mark the meal as “Served,” ensuring correct redemption
- Prevents misuse, reduces queue time, and provides accurate consumption data



Links:

- **GitHub PublicRepository**[https://github.com/ankush1902-vns/Mess Management TechFest](https://github.com/ankush1902-vns/Mess_Management_TechFest)
- **Demo Video Link :**
<https://drive.google.com/drive/folders/1LtWBmYYb4F15-LWZKz6EoN7wNrLTzc5T>
- **MVP Link :** <https://canteen-wise-buddy.lovable.app>